Practical No. 1 – DHCP Server Installation in Ubuntu

Aim:

1. To install the ISC DHCP server package in Ubuntu.
2. To enable the DHCP server service at system startup and verify its status.

Theory

The Dynamic Host Configuration Protocol (DHCP) is a network protocol used to automatically assign IP addresses and related configuration (subnet mask, gateway, DNS server, etc.) to devices on a network.
Without DHCP, network administrators must assign IPs manually to each device — which is time-consuming and prone to errors.

In Linux, one common implementation is the ISC DHCP Server package.
How it works:

1. A client device sends a DHCP Discover request on the network.
2. The DHCP server responds with a DHCP Offer — an available IP address and configuration details.
3. The client sends a DHCP Request to accept the offer.
4. The server sends a DHCP Acknowledgement, officially assigning the IP to the client.

Benefits of using DHCP:

- Automatic IP management
- Reduces IP conflicts
- Simplifies large network administration
- Supports automatic IP changes in dynamic environments

Requirements

- Ubuntu OS (20.04 / 22.04 recommended)
- Internet connection (for apt package installation)
- Sudo/root privileges
- Basic knowledge of Linux commands

Step-By-Step Procedure

Step 1: Install the ISC DHCP Server package

```
sudo apt install isc-dhcp-server -y
```

```
fycs@fycs-VirtualBox:~$ sudo apt install isc-dhcp-server -y
[sudo] password for fycs:
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  libirs-export141 libisccfg-export140
Suggested packages:
  isc-dhcp-server-ldap policycoreutils
```

- **Purpose**: Downloads and installs the ISC DHCP server software along with any dependencies.
- **Explanation:**
  - sudo → runs the command with admin rights.
  - apt install → installs packages from the Ubuntu repository.
  - isc-dhcp-server → name of the DHCP server package.
  - -y → automatically confirms installation.

Step 2: Verify package installation

dpkg -l | grep isc-dhcp-server

```
fycs@fycs-VirtualBox:~$ dpkg -l | grep isc-dhcp-server
ii  isc-dhcp-server                              4.3.3-5ubuntu12.10
                amd64        ISC DHCP server for automatic IP address assignment
```

Purpose: Checks if the DHCP server was installed successfully.

Explanation:

- dpkg -l → Lists all installed packages.
- grep isc-dhcp-server → Filters results to only show DHCP server package info.

Step 3: Enable service to start at boot

sudo systemctl enable isc-dhcp-server

```
fycs@fycs-VirtualBox:~$ sudo systemctl enable isc-dhcp-server
Synchronizing state of isc-dhcp-server.service with SysV init with /lib/systemd/
systemd-sysv-install...
Executing /lib/systemd/systemd-sysv-install enable isc-dhcp-server
```

- **Purpose:** Ensures the DHCP server starts automatically after system reboots.
- **Explanation:**
  - systemctl enable → Configures a service to launch on boot.
  - This creates a symbolic link from the system's startup sequence to the DHCP server's service file.

Step 4: Verify DHCP service status

systemctl status isc-dhcp-server

```
fycs@fycs-VirtualBox:~$ sudo systemctl status isc-dhcp-server
● isc-dhcp-server.service - ISC DHCP IPv4 server
   Loaded: loaded (/lib/systemd/system/isc-dhcp-server.service; enabled; vendor
   Active: failed (Result: exit-code) since Tue 2025-08-12 21:15:41 IST; 1min 30
     Docs: man:dhcpd(8)
 Main PID: 7113 (code=exited, status=1/FAILURE)

Aug 12 21:15:41 fycs-VirtualBox sh[7113]: If you think you have received this me
Aug 12 21:15:41 fycs-VirtualBox dhcpd[7113]: than a configuration issue please r
```

- Purpose: Displays the current state of the service.
- Explanation:
    - The status includes whether the service is active (running), inactive, or failed.

Observation / Output

- The ISC DHCP Server package was successfully installed.
- The service was enabled to start on boot and is currently running.
- Commands executed match the aim of the practical.

Conclusion

We have successfully installed and enabled the ISC DHCP server on an Ubuntu system. The service is active and ready to assign IP addresses to network clients automatically.

Troubleshooting / Notes

- Error: "E: Unable to locate package isc-dhcp-server" → Run sudo apt update before install.
- Error: Service failed to start → Check /etc/dhcp/dhcpd.conf syntax and ensure the network interface is configured.

Practical No. 2 – IPv6 & Network Configuration in Ubuntu

Aim

1. Add a new user and configure network settings.
2. Change network to a static IP.
3. Disable IPv6 if not required.
4. Manage and configure system services.
5. Stop and disable unwanted services and configure sudo.

Theory

In Linux, network configuration is essential for communication and resource sharing.

- IPv6 is the latest Internet Protocol version, offering a huge address space. It can be disabled if the network doesn't support it to reduce complexity.
- Static IP ensures a fixed address for servers/services.
- systemctl is used to manage services — start, stop, enable, disable.
- User management allows control over system access.

Requirements

- Ubuntu OS (20.04/22.04)
- Sudo privileges
- Internet connection
- Basic knowledge of network commands (ifconfig, netplan, systemctl)

Step-by-Step Procedure

Step 1 – Create a new user

sudo useradd Tycs

sudo passwd Tycs

id Tycs

sudo apt install net-tools

```
fycs@fycs-VirtualBox:~$ sudo useradd tycs
fycs@fycs-VirtualBox:~$ sudo passwd tycs
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
fycs@fycs-VirtualBox:~$ id tycs
uid=1004(tycs) gid=1004(tycs) groups=1004(tycs)
fycs@fycs-VirtualBox:~$ sudo apt install net-tools
Reading package lists... Done
Building dependency tree
Reading state information... Done
net-tools is already the newest version (1.60-26ubuntu1).
```

Adds a user named Tycs and verifies creation. Installs net-tools for network commands.

Step 2 – Check network interface

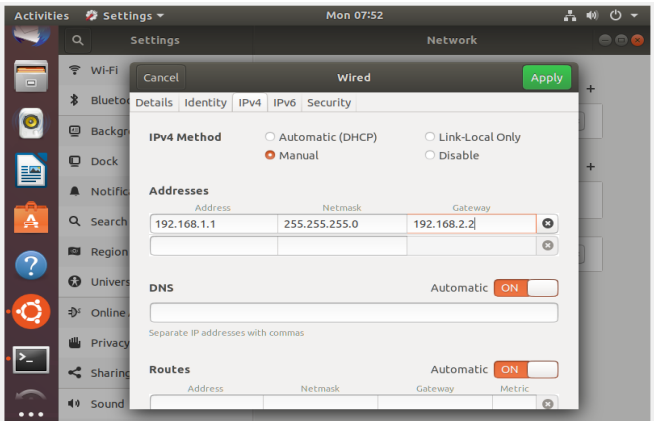
Ifconfig

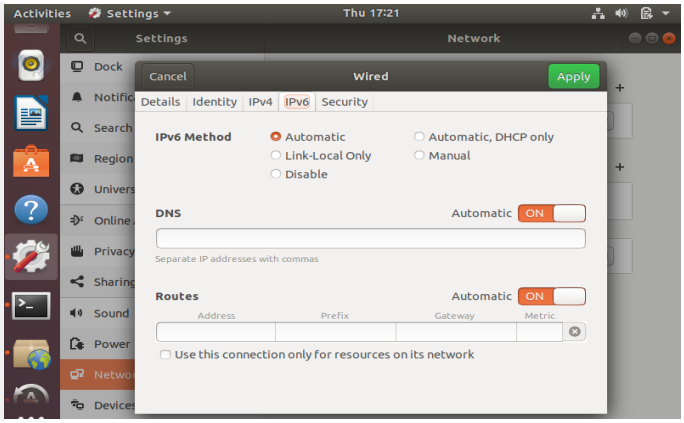
```
fycs@fycs-VirtualBox:~$ ifconfig
enp0s3    Link encap:Ethernet  HWaddr 08:00:27:86:20:b1
          inet addr:10.0.2.15  Bcast:10.0.2.255  Mask:255.255.255.0
          inet6 addr: fe80::28b:9420:82eb:aed2/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:2660 errors:0 dropped:0 overruns:0 frame:0
          TX packets:1550 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:2918697 (2.9 MB)  TX bytes:119282 (119.2 KB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:1093 errors:0 dropped:0 overruns:0 frame:0
          TX packets:1093 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:99404 (99.4 KB)  TX bytes:99404 (99.4 KB)
```

Displays available interfaces and IP configuration.

Step 3 – Configure Static IP & Disable IPv6 (if needed)

- Edit network settings via GUI





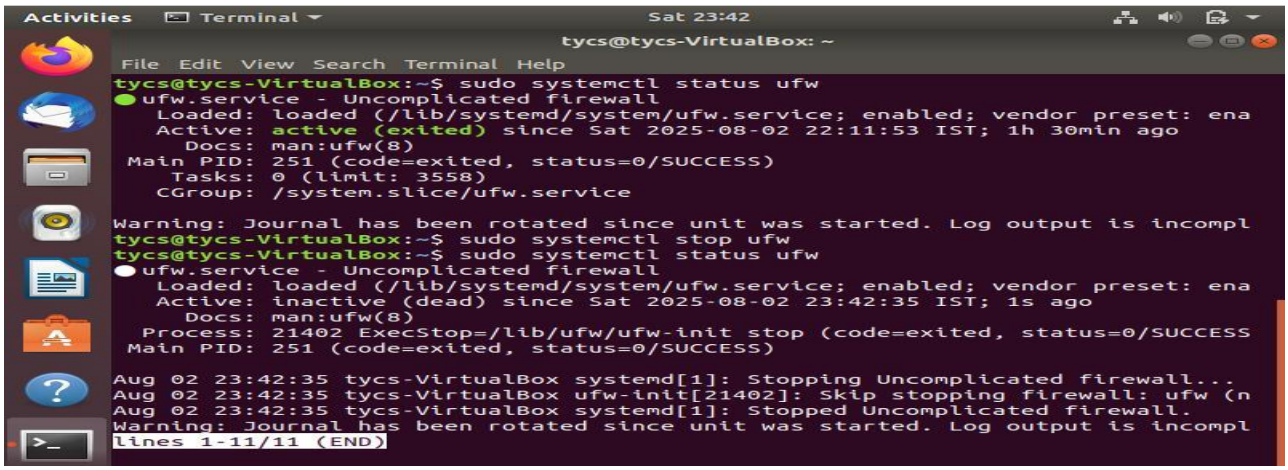
Step 4 – Manage a service (Example: UFW Firewall)

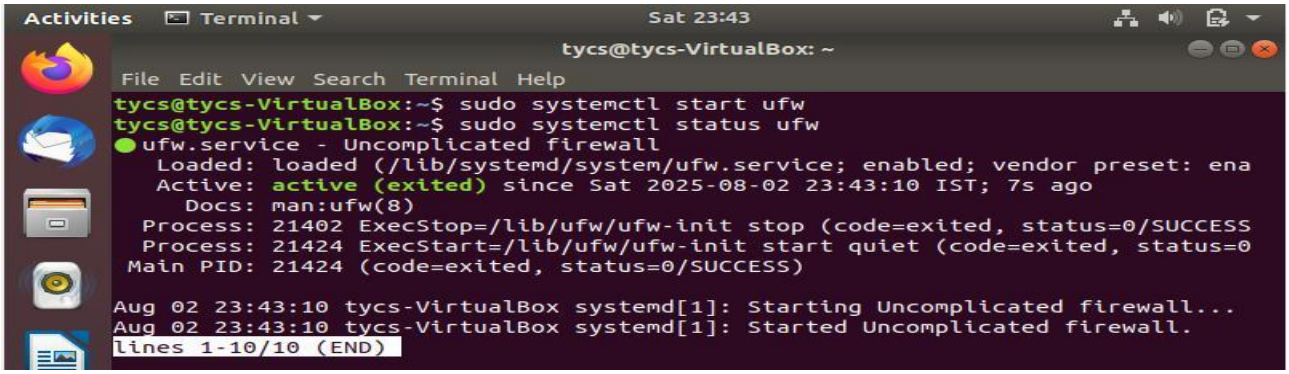
sudo systemctl status ufw

sudo systemctl stop ufw

sudo systemctl disable ufw

sudo systemctl status ufw





Stops and disables firewall service.

Observation / Output

- New user created.
- Network switched to static IP.
- IPv6 disabled (if not needed).
- Listed and stopped unwanted services.

Conclusion

Successfully created a new user, configured network settings, managed IP configuration, and controlled running services as per requirements.

Notes / Troubleshooting

- If ifconfig not found → run sudo apt install net-tools.
- Always backup netplan files before editing.
- Ensure service disable does not break essential networking.

Practical No. 3A – NTPD Server (America Timezone) Configuration

Aim

1. Install and configure an NTP (Network Time Protocol) server.
2. Configure NTP clients on Ubuntu and Windows.
3. Synchronize system time with an America-based timezone.

Theory

The Network Time Protocol (NTP) is used to synchronize the system clock of computers over a network.

- NTP Server provides accurate time.
- NTP Client adjusts its local clock based on the server's time.
- Accurate time is critical in networking for logging, security, and scheduling.

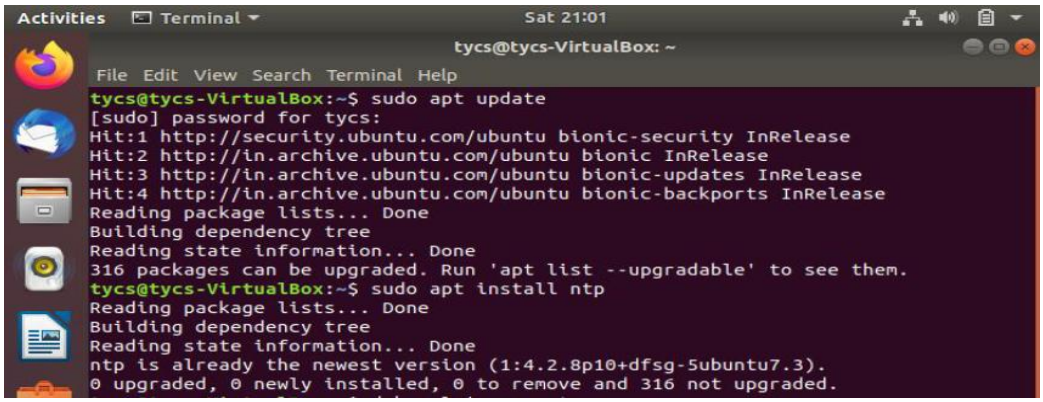
The ntpd service in Linux is the most common implementation.

Requirements

- Ubuntu OS with internet access.
- Sudo privileges.
- Basic understanding of Linux service management.

Step-by-Step Procedure

Step 1 – Update and install NTP



Installs the NTP server software.

Step 2 – Verify installation

dpkg -l | grep ntp

sudo systemctl status ntp

```
tycs@tycs-VirtualBox:~$ dpkg -l | grep ntp
ii  ntp                                  1:4.2.8p10+dfsg-5ubuntu7.3
                amd64        Network Time Protocol daemon and utility programs
ii  sntp                                 1:4.2.8p10+dfsg-5ubuntu7.3
                amd64        Network Time Protocol - sntp client
```

```
Activities  Terminal  Sat 11:52
tycs@tycs-VirtualBox: ~
File  Edit  View  Search  Terminal  Help
tycs@tycs-VirtualBox:~$ sudo systemctl status ntp
● ntp.service - Network Time Service
   Loaded: loaded (/lib/systemd/system/ntp.service; enabled; vendor preset: ena
   Active: active (running) since Sat 2025-08-02 11:44:00 EDT; 6min ago
     Docs: man:ntpd(8)
  Process: 3869 ExecStart=/usr/lib/ntp/ntp-systemd-wrapper (code=exited, status
 Main PID: 3877 (ntpd)
    Tasks: 2 (limit: 3558)
   CGroup: /system.slice/ntp.service
           └─3877 /usr/sbin/ntpd -p /var/run/ntpd.pid -g -u 123:129
```

Checks if package is installed and service is running**.**

Step 3 – Edit NTP configuration

sudo nano /etc/ntp.conf

```
Activities  Terminal  Sat 11:51
tycs@tycs-VirtualBox: ~
File  Edit  View  Search  Terminal  Help
  GNU nano 2.9.3                /etc/ntp.conf                    Modified

filegen peerstats file peerstats type day enable
filegen clockstats file clockstats type day enable

# Specify one or more NTP servers.

# Use servers from the NTP Pool Project. Approved by Ubuntu Technical Board
# on 2011-02-08 (LP: #104525). See http://www.pool.ntp.org/join.html for
# more information.
pool 0.ubuntu.pool.ntp.org iburst
pool 1.ubuntu.pool.ntp.org iburst
pool 2.ubuntu.pool.ntp.org iburst
pool 3.ubuntu.pool.ntp.org iburst

# Use Ubuntu's ntp server as a fallback.
pool ntp.ubuntu.com

# Access control configuration; see /usr/share/doc/ntp-doc/html/accopt.html for
# details.  The web page <http://support.ntp.org/bin/view/Support/AccessRestri$
# might also be helpful.
#
# Note that "restrict" applies to both servers and clients, so a configuration
# that might be intended to block requests from certain clients could also end
# up blocking replies from your own upstream servers.

^G Get Help   ^O Write Out   ^W Where Is   ^K Cut Text   ^J Justify
^X Exit       ^R Read File   ^\ Replace    ^U Uncut Text ^T To Spell
```

```
Activities   Terminal   Sat 21:07
tycs@tycs-VirtualBox: ~
File  Edit  View  Search  Terminal  Help
  GNU nano 2.9.3                /etc/ntp.conf                Modified

filegen peerstats file peerstats type day enable
filegen clockstats file clockstats type day enable

# Specify one or more NTP servers.

# Use servers from the NTP Pool Project. Approved by Ubuntu Technical Board
# on 2011-02-08 (LP: #104525). See http://www.pool.ntp.org/join.html for
# more information.
pool 0.in.pool.ntp.org iburst
pool 1.in.pool.ntp.org iburst
pool 2.in.pool.ntp.org iburst
pool 3.in.pool.ntp.org iburst

# Use Ubuntu's ntp server as a fallback.
pool ntp.ubuntu.com

# Access control configuration; see /usr/share/doc/ntp-doc/html/accopt.html for
# details.  The web page <http://support.ntp.org/bin/view/Support/AccessRestri$
# might also be helpful.
#
# Note that "restrict" applies to both servers and clients, so a configuration
# that might be intended to block requests from certain clients could also end
# up blocking replies from your own upstream servers.

^G Get Help    ^O Write Out   ^W Where Is    ^K Cut Text    ^J Justify
^X Exit        ^R Read File   ^\ Replace     ^U Uncut Text  ^T To Spell
```

Modify server pool entries if needed.

Step 4 – Enable and start NTP service

sudo systemctl start ntp

sudo systemctl status ntp

```
tycs@tycs-VirtualBox:~$ sudo systemctl start ntp
tycs@tycs-VirtualBox:~$ sudo systemctl status ntp
● ntp.service - Network Time Service
   Loaded: loaded (/lib/systemd/system/ntp.service; enabled; vendor preset: ena
   Active: active (running) since Sat 2025-08-02 21:29:41 IST; 4min 32s ago
     Docs: man:ntpd(8)
  Process: 5632 ExecStart=/usr/lib/ntp/ntp-systemd-wrapper (code=exited, status
 Main PID: 5640 (ntpd)
    Tasks: 2 (limit: 3558)
   CGroup: /system.slice/ntp.service
           └─5640 /usr/sbin/ntpd -p /var/run/ntpd.pid -g -u 123:129

Aug 02 21:29:46 tycs-VirtualBox ntpd[5640]: Soliciting pool server 172.105.60.1
Aug 02 21:29:46 tycs-VirtualBox ntpd[5640]: Soliciting pool server 192.46.210.3
Aug 02 21:29:46 tycs-VirtualBox ntpd[5640]: Soliciting pool server 14.139.60.10
Aug 02 21:29:46 tycs-VirtualBox ntpd[5640]: Soliciting pool server 91.189.91.15
Aug 02 21:29:47 tycs-VirtualBox ntpd[5640]: Soliciting pool server 185.125.190.
Aug 02 21:29:47 tycs-VirtualBox ntpd[5640]: Soliciting pool server 162.159.200.
Aug 02 21:29:47 tycs-VirtualBox ntpd[5640]: Soliciting pool server 86.107.100.2
Aug 02 21:29:48 tycs-VirtualBox ntpd[5640]: Soliciting pool server 185.125.190.
Aug 02 21:29:48 tycs-VirtualBox ntpd[5640]: Soliciting pool server 139.84.142.1
Aug 02 21:29:48 tycs-VirtualBox ntpd[5640]: Soliciting pool server 2401:c080:30
tycs@tycs-VirtualBox:~$
```

Step 5 – Check current peer status

ntpq -p

```
tycs@tycs-VirtualBox:~$ ntpq -p
     remote           refid      st t when poll reach   delay   offset  jitter
==============================================================================
 0.in.pool.ntp.o .POOL.          16 p    -   64    0    0.000    0.000   0.000
 1.in.pool.ntp.o .POOL.          16 p    -   64    0    0.000    0.000   0.000
 2.in.pool.ntp.o .POOL.          16 p    -   64    0    0.000    0.000   0.000
 3.in.pool.ntp.o .POOL.          16 p    -   64    0    0.000    0.000   0.000
 ntp.ubuntu.com  .POOL.          16 p    -   64    0    0.000    0.000   0.000
+103.189.191.36  14.139.60.107    2 u    4   64    1   26.041    1.204   8.772
-ntp1.ggsrv.de   134.71.66.21     2 u    5   64    1  162.020   -0.044   1.673
-103.189.191.15  14.139.60.107    2 u    4   64    1   34.746   -2.943  12.181
+ntp6.mum-in.hos 193.79.237.14    2 u    2   64    1   21.909    2.184   3.317
-ec2-3-111-45-10 169.254.169.123  4 u    5   64    1    8.821   -1.534   4.972
#time.cloudflare 10.165.8.4       3 u    4   64    1    9.802  -17.019   4.743
+157.245.102.2   129.134.27.123   2 u    1   64    1   17.211    2.283   3.402
-160.191.28.135  3.111.45.100     5 u    2   64    1    9.797   -0.638   2.123
#ntp-pool.time.4 139.59.15.185    6 u    2   64    1    9.538   -6.245   2.295
-ntp7.mum-in.hos 193.79.237.14    2 u    3   64    1    8.792    4.690   8.198
```

Verifies NTP peers.

Step 6 – Check and set timezone to America/Santiago

timedatectl list-timezones | grep America

sudo timedatectl set-timezone America/Santiago

```
tycs@tycs-VirtualBox:~$ timedatectl
                      Local time: Sat 2025-08-02 21:30:14 IST
                  Universal time: Sat 2025-08-02 16:00:14 UTC
                        RTC time: Sat 2025-08-02 16:00:15
                       Time zone: Asia/Kolkata (IST, +0530)
       System clock synchronized: yes
systemd-timesyncd.service active: yes
                 RTC in local TZ: no
tycs@tycs-VirtualBox:~$ date
Sat Aug  2 21:30:22 IST 2025
tycs@tycs-VirtualBox:~$
```

```
tycs@tycs-VirtualBox:~$ timedatectl list-timezones | grep America
America/Adak
America/Anchorage
America/Anguilla
America/Antigua
America/Araguaina
America/Argentina/Buenos_Aires
America/Argentina/Catamarca
America/Argentina/Cordoba
America/Argentina/Jujuy
America/Argentina/La_Rioja
America/Argentina/Mendoza
America/Argentina/Rio_Gallegos
America/Argentina/Salta
America/Argentina/San_Juan
```

```
Activities  Terminal  Sat 11:56
tycs@tycs-VirtualBox: ~
File  Edit  View  Search  Terminal  Help
America/Whitehorse
America/Winnipeg
America/Yakutat
America/Yellowknife
tycs@tycs-VirtualBox:~$ sudo timedatectl set-timezone America/New_York
tycs@tycs-VirtualBox:~$ sudo systemctl restart ntp
tycs@tycs-VirtualBox:~$ ntpq -p
     remote           refid      st t when poll reach   delay   offset  jitter
==============================================================================
 0.us.pool.ntp.o .POOL.          16 p    -   64    0    0.000    0.000   0.000
 1.us.pool.ntp.o .POOL.          16 p    -   64    0    0.000    0.000   0.000
 2.us.pool.ntp.o .POOL.          16 p    -   64    0    0.000    0.000   0.000
 3.us.pool.ntp.o .POOL.          16 p    -   64    0    0.000    0.000   0.000
 ntp.ubuntu.com  .POOL.          16 p    -   64    0    0.000    0.000   0.000
 204.2.134.172   22.42.17.250     3 u    8   64    1 1412.83 -564.37 834.368
 108.61.215.221  158.51.99.19     3 u   12   64    1 1467.45 -607.93 608.157
 ntp.maxhost.io  132.163.97.4     2 u    1   64    1  254.225   -7.162  58.720
tycs@tycs-VirtualBox:~$ timedatectl
                      Local time: Sat 2025-08-02 11:56:39 EDT
                  Universal time: Sat 2025-08-02 15:56:39 UTC
                        RTC time: Sat 2025-08-02 15:56:40
                       Time zone: America/New_York (EDT, -0400)
       System clock synchronized: yes
systemd-timesyncd.service active: yes
                 RTC in local TZ: no
tycs@tycs-VirtualBox:~$ date
Sat Aug  2 11:56:48 EDT 2025
tycs@tycs-VirtualBox:~$
```

Observation / Output

- **NTP server installed and running.**

- **Timezone changed to America/Santiago.**
- **Time synchronized with remote NTP peers.**

Conclusion

Successfully installed and configured NTPD server, synchronized time with an American timezone, and verified service operation.

Notes / Troubleshooting

- **Use timedatectl status to check if NTP sync is active.**
- **If firewall is active, allow UDP port 123 for NTP.**

Practical No. 3B – NTPD Server (India Timezone) Configuration

Aim

1. Configure the NTP server for Indian Standard Time (IST).
2. Set up an NTP client (Ubuntu & Windows) to synchronize with the server.
3. Verify time synchronization.

Theory

The Network Time Protocol (NTP) keeps computers' clocks synchronized over a network.

- An NTP Server distributes the correct time to clients.
- The ntpd service ensures that time updates happen automatically.
- Accurate time is crucial for system logs, security, and coordination between servers.

In this practical, we extend the NTP setup from Practical 3A by changing the timezone to Asia/Kolkata (India) and ensuring clients sync correctly.

Requirements

- Ubuntu server with NTP installed (from Practical 3A).
- Sudo privileges.
- Network access for client systems.

Step-by-Step Procedure

Step 1 – Check current date/time

```
fycs@fycs-VirtualBox:~$ date
Wed Aug 13 02:46:20 IST 2025
```

Shows system time before changes.

Step 2 – List available Indian timezones

timedatectl list-timezones | grep India

```
fycs@fycs-VirtualBox:~$ timedatectl list-timezones |grep Indian
America/Indiana/Indianapolis
America/Indiana/Knox
America/Indiana/Marengo
America/Indiana/Petersburg
America/Indiana/Tell_City
America/Indiana/Vevay
America/Indiana/Vincennes
America/Indiana/Winamac
Indian/Antananarivo
Indian/Chagos
```

Step 3 – Ensure NTP service is enabled & running

sudo systemctl start ntp

sudo systemctl status ntp

```
tycs@tycs-VirtualBox:~$ sudo systemctl start ntp
tycs@tycs-VirtualBox:~$ sudo systemctl status ntp
● ntp.service - Network Time Service
   Loaded: loaded (/lib/systemd/system/ntp.service; enabled; vendor preset: ena
   Active: active (running) since Sat 2025-08-02 21:29:41 IST; 4min 32s ago
     Docs: man:ntpd(8)
  Process: 5632 ExecStart=/usr/lib/ntp/ntp-systemd-wrapper (code=exited, status
 Main PID: 5640 (ntpd)
    Tasks: 2 (limit: 3558)
   CGroup: /system.slice/ntp.service
           └─5640 /usr/sbin/ntpd -p /var/run/ntpd.pid -g -u 123:129
```

Step 4 – Set timezone to India

sudo timedatectl set-timezone Asia/Kolkata

Step 5 – Verify synchronization

sudo timedatectl

```
fycs@fycs-VirtualBox:~$ sudo timedatectl set-timezone Asia/Kolkata
[sudo] password for fycs:
fycs@fycs-VirtualBox:~$ timedatectl
      Local time: Wed 2025-08-13 02:55:07 IST
  Universal time: Tue 2025-08-12 21:25:07 UTC
        RTC time: Tue 2025-08-12 21:25:08
       Time zone: Asia/Kolkata (IST, +0530)
 Network time on: yes
NTP synchronized: yes
 RTC in local TZ: no
```

Observation / Output

- Timezone successfully set to Asia/Kolkata (IST).
- NTP service is active and synchronizing with configured servers.
- Time on clients matches the NTP server.

Conclusion

We successfully reconfigured the NTPD server to use Indian Standard Time, enabled service synchronization, and verified correct time updates across clients.

Notes / Troubleshooting

- If sync fails, check /etc/ntp.conf server entries.
- Allow UDP port 123 if firewall is active.

Practical No. 4 – SSH Server & Client Configuration

Aim

1. Install and configure SSH server for secure remote login.
2. Configure SSH client connections (Ubuntu & Windows).
3. Manage SSH service authentication and firewall settings.

Theory

SSH (Secure Shell) is a protocol that allows secure remote login and command execution over an encrypted channel.

- SSH Server runs on the host to accept connections.
- SSH Client connects to the server using username/password or keys.
- Common use cases: remote administration, file transfer (SCP/SFTP), tunneling.
- In Windows, tools like PuTTY can connect to Linux SSH servers.

Requirements

- Ubuntu server with internet access.
- Sudo privileges.
- SSH client (Linux terminal or PuTTY for Windows).

Step by step procedure

Step 1 – Check network interface (Get server IP)

Ifconfig

```
fycs@fycs-VirtualBox:~$ ifconfig
enp0s3    Link encap:Ethernet  HWaddr 08:00:27:86:20:b1
          inet addr:10.0.2.15  Bcast:10.0.2.255  Mask:255.255.255.0
          inet6 addr: fe80::28b:9420:82eb:aed2/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:2660 errors:0 dropped:0 overruns:0 frame:0
          TX packets:1550 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:2918697 (2.9 MB)  TX bytes:119282 (119.2 KB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:1093 errors:0 dropped:0 overruns:0 frame:0
          TX packets:1093 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:99404 (99.4 KB)  TX bytes:99404 (99.4 KB)
```

Note the IP address for connecting via SSH.

Step 2 – Install SSH server

sudo apt install openssh-server -y

sudo systemctl enable ssh

sudo systemctl start ssh

sudo systemctl status ssh

```
fycs@fycs-VirtualBox:~$ sudo apt install openssh-server
[sudo] password for fycs:
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  ncurses-term openssh-sftp-server ssh-import-id
Suggested packages:
  ssh-askpass rssh molly-guard monkeysphere
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
0 upgraded, 4 newly installed, 0 to remove and 167 not upgraded.
Need to get 633 kB of archives.
```

```
tycs@tycs-VirtualBox:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /lib/systemd/s
ystemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable ssh
tycs@tycs-VirtualBox:~$ sudo systemctl start ssh
tycs@tycs-VirtualBox:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: ena
   Active: active (running) since Sun 2025-08-10 09:16:36 IST; 22min ago
 Main PID: 677 (sshd)
    Tasks: 1 (limit: 3558)
   CGroup: /system.slice/ssh.service
           └─677 /usr/sbin/sshd -D

Aug 10 09:35:00 tycs-VirtualBox systemd[1]: Reloading OpenBSD Secure Shell serv
Aug 10 09:35:00 tycs-VirtualBox sshd[677]: Received SIGHUP; restarting.
Aug 10 09:35:00 tycs-VirtualBox systemd[1]: Reloaded OpenBSD Secure Shell serve
Aug 10 09:35:00 tycs-VirtualBox sshd[677]: Server listening on 0.0.0.0 port 22.
Aug 10 09:35:00 tycs-VirtualBox sshd[677]: Server listening on :: port 22.
Aug 10 09:36:19 tycs-VirtualBox systemd[1]: Reloading OpenBSD Secure Shell serv
Aug 10 09:36:19 tycs-VirtualBox sshd[677]: Received SIGHUP; restarting.
Aug 10 09:36:19 tycs-VirtualBox systemd[1]: Reloaded OpenBSD Secure Shell serve
Aug 10 09:36:19 tycs-VirtualBox sshd[677]: Server listening on 0.0.0.0 port 22.
Aug 10 09:36:19 tycs-VirtualBox sshd[677]: Server listening on :: port 22.
```

Step-2: Create a local test directory and file

mkdir Jayad

cd Jayad

nano J1.txt

```
fycs@fycs-VirtualBox:~$ mkdir Jayad
fycs@fycs-VirtualBox:~$ cd Jayad
fycs@fycs-VirtualBox:~/Jayad$ ls
fycs@fycs-VirtualBox:~/Jayad$ cd..
cd..: command not found
fycs@fycs-VirtualBox:~/Jayad$ cd ..
fycs@fycs-VirtualBox:~$ cd Jayad
fycs@fycs-VirtualBox:~/Jayad$ nano J1.txt
```

```
fycs@fycs-VirtualBox: ~/Jayad
  GNU nano 2.5.3                File: J1.txt

Hello this is openssh practical

                              [ Read 1 line ]
^G Get Help  ^O Write Out ^W Where Is  ^K Cut Text  ^J Justify   ^C Cur Pos
^X Exit      ^R Read File ^\ Replace  ^U Uncut Text^T To Spell  ^_ Go To Line
```

Type the following inside nano editor:

Save and exit nano (Ctrl + O, Enter, Ctrl + X). Then list the files:

ls

```
fycs@fycs-VirtualBox:~/Jayad$ ls
J1.txt
```

Step-3: Get server IP and configure PuTTY on Windows

```
tycs@tycs-VirtualBox:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.3.20  netmask 255.255.255.0  broadcast 192.168.3.255
        inet6 fe80::a00:27ff:fec9:3ede  prefixlen 64  scopeid 0x20<link>
        inet6 2409:4081:1d00:d80:a00:27ff:fec9:3ede  prefixlen 64  scopeid 0x0<
global>
        ether 08:00:27:c9:3e:de  txqueuelen 1000  (Ethernet)
        RX packets 222  bytes 26503 (26.5 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 328  bytes 42102 (42.1 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
```

Note down the IP address (e.g., 192.168.3.20).

On Windows:

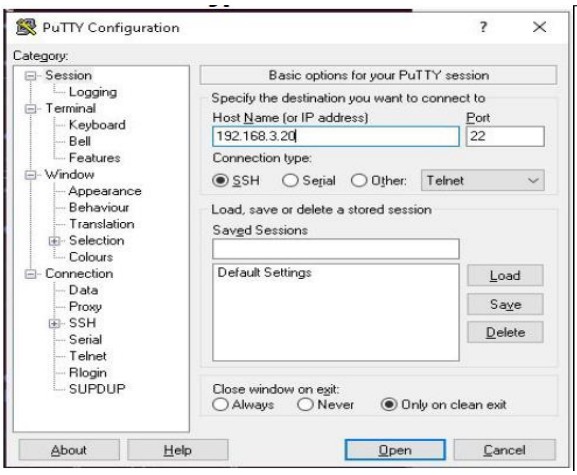
- Download and install PuTTY.
- Open PuTTY.
- In "Host Name", enter the IP address noted above.
- Set Port to 22.
- Select "SSH" as connection type.
- Click "Open" and accept any security warnings.

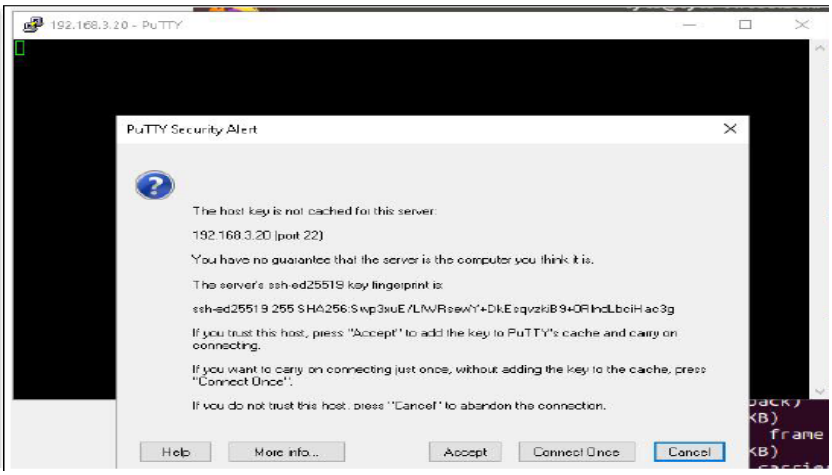
Step-4: Login via SSH and perform remote commands
Log in with your Ubuntu username and password:
Username: tycs

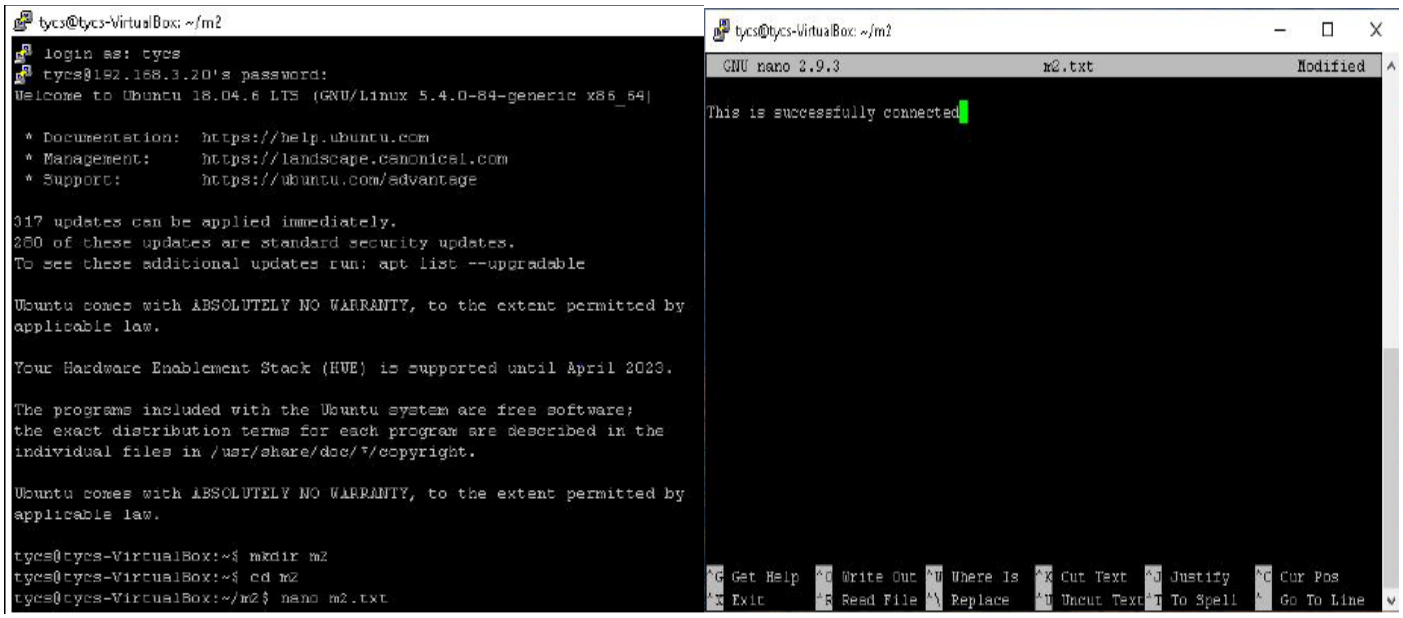
Password: [your password]
After successful login, run:
mkdir m2

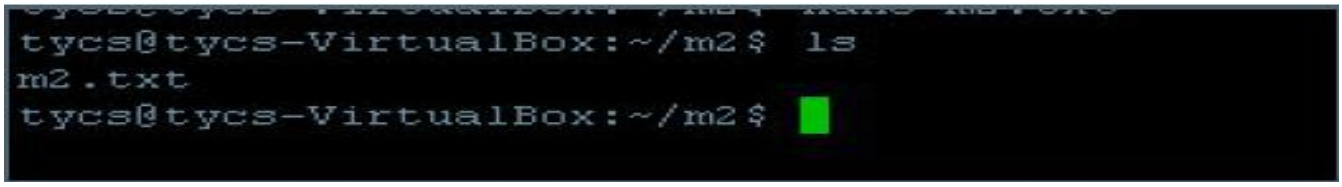
cd m2

nano m2.txt
Inside nano, type:
This is successfully connected
Save and exit, then list files:
ls









Observation / Output

- OpenSSH server installed and running.
- Able to connect remotely using PuTTY with correct credentials.
- Successfully created and edited files remotely through the SSH session.
- Verified network interface and IP address for connection.

Conclusion

The OpenSSH server was installed on Ubuntu, and remote connection through SSH was successfully established from a Windows client using PuTTY. Basic file operations were performed remotely, demonstrating secure and functional SSH access.

Practical No. 5 – Configure DHCP Server

Aim

- To configure and manage the Dynamic Host Configuration Protocol (DHCP) server on Ubuntu.
- To setup DHCP to dynamically assign IP addresses to client machines on the network.

Theory

DHCP (Dynamic Host Configuration Protocol) is a network management protocol used on IP networks, where a DHCP server automatically assigns IP addresses and other communication parameters to devices (clients) connected to the network. This enables them to communicate efficiently without manual network configuration.

The DHCP process involves:

- Clients sending a DHCP Discover request to find available DHCP servers.
- Servers responding with a DHCP Offer containing IP configuration information.
- Clients sending a DHCP Request to the selected server.
- Server sends DHCP Acknowledgement, confirming the lease of the IP address.

This automatic IP assignment reduces manual errors, simplifies network management, and supports dynamic networks.

Requirements

- Ubuntu system with the DHCP server package installed (isc-dhcp-server).
- Sudo privileges for administrative commands.
- Network interface with a static IP configured (required for DHCP server).

Step-by-Step Procedure

Note: This practical assumes the DHCP server is already installed. If not, install it first using:

Step-1: Check network interface and configure DHCP server to use it

ifconfig

sudo nano /etc/default/isc-dhcp-server

```
tycs@tycs-VirtualBox:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.3.20  netmask 255.255.255.0  broadcast 192.168.3.255
        inet6 fe80::a00:27ff:fec9:3ede  prefixlen 64  scopeid 0x20<link>
        inet6 2409:4081:1d00:d80:a00:27ff:fec9:3ede  prefixlen 64  scopeid 0x0<
global>
        ether 08:00:27:c9:3e:de  txqueuelen 1000  (Ethernet)
        RX packets 222  bytes 26503 (26.5 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 328  bytes 42102 (42.1 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
```

```
tycs@tycs-VirtualBox: /etc/dhcp
File Edit View Search Terminal Help
tycs@tycs-VirtualBox:~$ sudo nano /etc/default/isc-dhcp-server
[sudo] password for tycs:
```

In the opened file, set your network interface (e.g. enp0s3) on the line:
INTERFACESv4="enp0s3"

```
tycs@tycs-VirtualBox: ~
File Edit View Search Terminal Help
  GNU nano 2.9.3              /etc/default/isc-dhcp-server

# Defaults for isc-dhcp-server (sourced by /etc/init.d/isc-dhcp-server)

# Path to dhcpd's config file (default: /etc/dhcp/dhcpd.conf).
#DHCPDv4_CONF=/etc/dhcp/dhcpd.conf
#DHCPDv6_CONF=/etc/dhcp/dhcpd6.conf

# Path to dhcpd's PID file (default: /var/run/dhcpd.pid).
#DHCPDv4_PID=/var/run/dhcpd.pid
#DHCPDv6_PID=/var/run/dhcpd6.pid

# Additional options to start dhcpd with.
#       Don't use options -cf or -pf here; use DHCPD_CONF/ DHCPD_PID instead
#OPTIONS=""

# On what interfaces should the DHCP server (dhcpd) serve DHCP requests?
#       Separate multiple interfaces with spaces, e.g. "eth0 eth1".
INTERFACESv4="enp0s3"
INTERFACESv6=""

                                        [ Read 18 lines ]
^G Get Help    ^O Write Out   ^W Where Is    ^K Cut Text    ^J Justify
^X Exit        ^R Read File   ^\ Replace     ^U Uncut Text  ^T To Spell
```

Step-2: Edit main DHCP configuration file

cd /etc/dhcp

sudo nano dhcpd.conf

```
tycs@tycs-VirtualBox:~$ cd /etc/dhcp
tycs@tycs-VirtualBox:/etc/dhcp$ sudo nano dhcpd.conf
```

In dhcpd.conf file, perform these important edits:

- Comment out these lines by adding # at the start:

  #option domain-name "example.org";

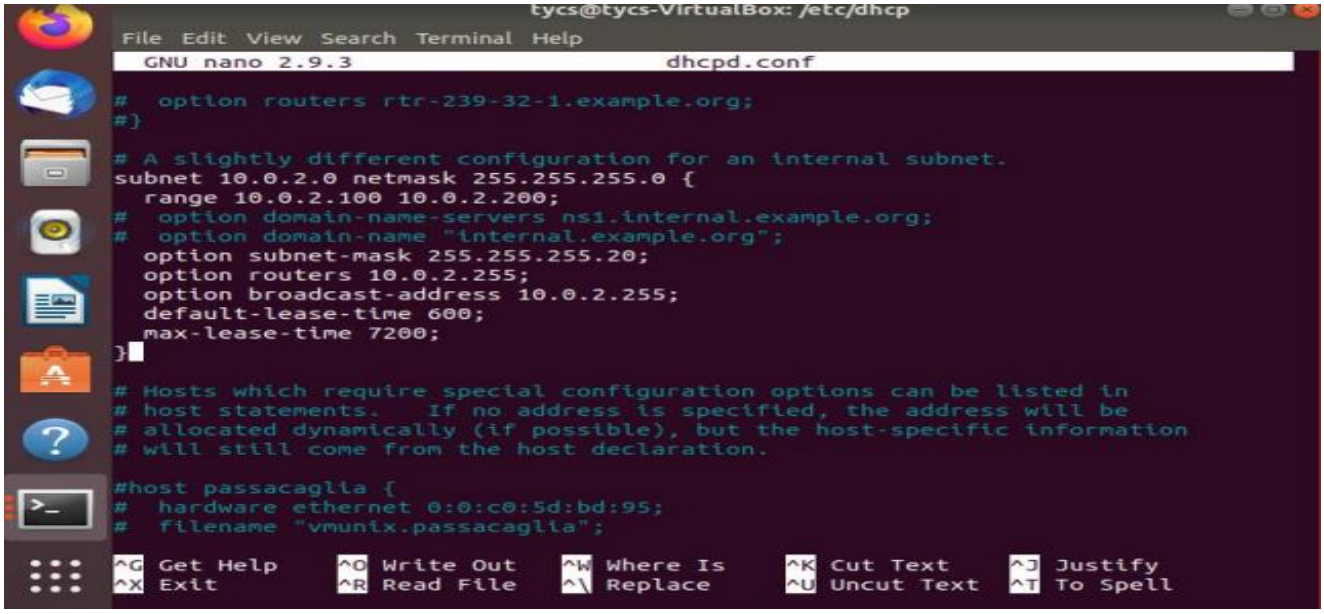
  #option domain-name-servers ns1.example.org;

  Uncomment authoritative by removing #:
  authoritative;

```
tycs@tycs-VirtualBox: /etc/dhcp
File Edit View Search Terminal Help
  GNU nano 2.9.3                    dhcpd.conf

dhcpd.conf
#
# Sample configuration file for ISC dhcpd
#
# Attention: If /etc/ltsp/dhcpd.conf exists, that will be used as
# configuration file instead of this file.
#

# option definitions common to all supported networks...
#option domain-name "example.org";
#option domain-name-servers ns1.example.org, ns2.example.org;

default-lease-time 600;
max-lease-time 7200;

# The ddns-updates-style parameter controls whether or not the server will
# attempt to do a DNS update when a lease is confirmed. We default to the
# behavior of the version 2 packages ('none', since DHCP v2 didn't
# have support for DDNS.)
ddns-update-style none;

# If this DHCP server is the official DHCP server for the local
# network, the authoritative directive should be uncommented.
authoritative;

# Use this to send dhcp log messages to a different log file (you also
# have to hack syslog.conf to complete the redirection).
#log-facility local7;

^G Get Help    ^O Write Out   ^W Where Is    ^K Cut Text    ^J Justify     ^C Cur Pos     M-U Undo
^X Exit        ^R Read File   ^\ Replace     ^U Uncut Text  ^T To Spell    ^_ Go To Line  M-E Redo
```

Uncomment and configure your subnet block as per your network, example:

subnet 10.0.2.0 netmask 255.255.255.0 { range 10.0.2.100 10.0.2.200; option subnet-mask 255.255.255.0; option routers 10.0.2.1; option broadcast-address 10.0.2.255; default-lease-time 600; max-lease-time 7200; }

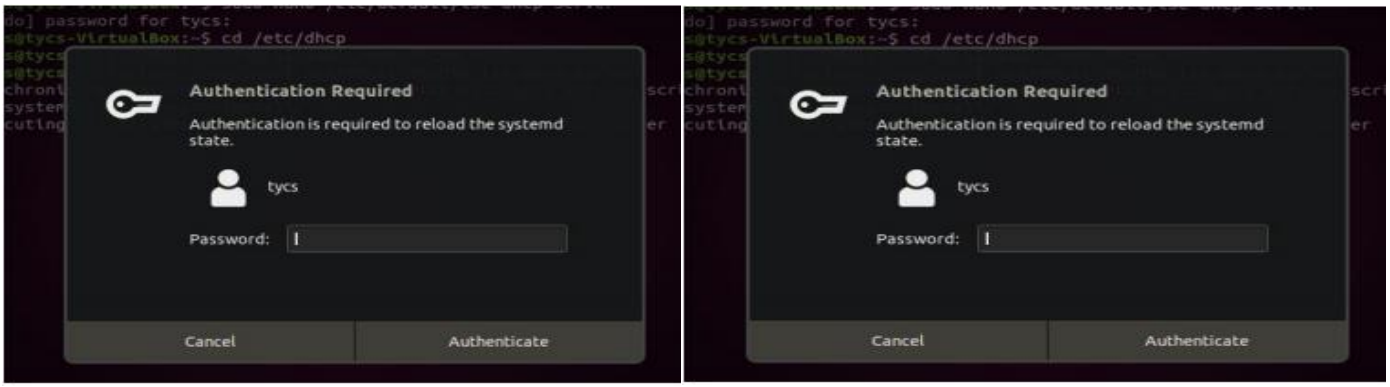


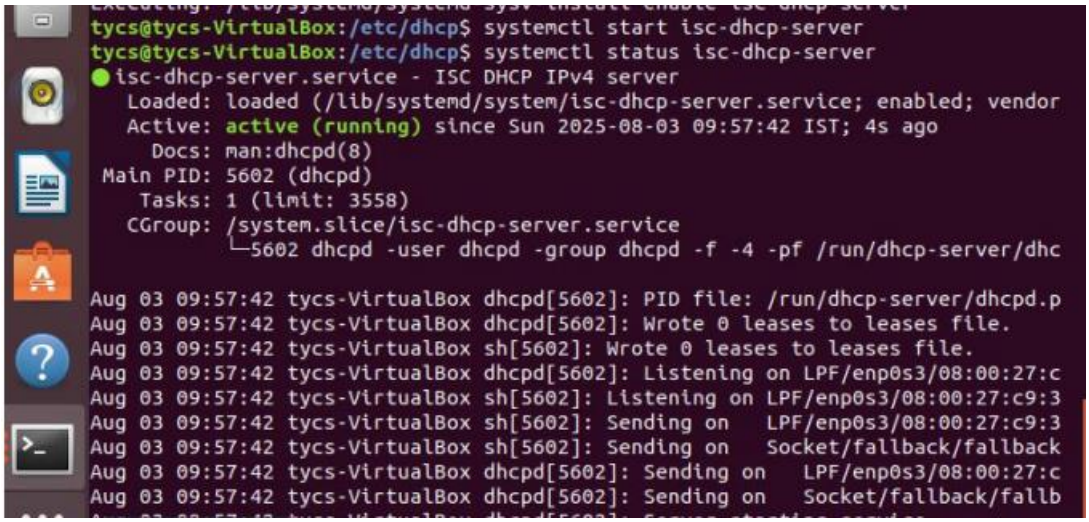
Step-3: Enable and start DHCP server services

sudo systemctl enable isc-dhcp-server

sudo systemctl start isc-dhcp-server

sudo systemctl status isc-dhcp-server





- You may be prompted for your password multiple times during service enable/start.

Observation / Output

- DHCP server is enabled on the chosen interface.
- DHCP service started and runs without errors.
- Clients on the network will receive dynamic IP addresses as per the configured IP range in the subnet.

Conclusion

The ISC DHCP server was successfully configured with a defined subnet range and network interface. The service is active and ready to assign IP addresses dynamically to clients in the network.

Troubleshooting / Notes

- If the server fails to start, check the syntax in /etc/dhcp/dhcpd.conf for errors.
- Confirm the network interface name matches the one on your system (ifconfig or ip a).
- Make sure the server has a static IP address configured on the DHCP-serving interface.
- Verify no other DHCP servers are running in the network to avoid conflicts.
- For firewall-enabled systems, allow DHCP ports UDP 67 and 68:

Practical No. 6 – Configure NFS Server and Client

Aim

- To configure the Network File System (NFS) server to share directories on your Ubuntu system.
- To configure NFS clients on Ubuntu and Windows to access shared directories.

Theory

The Network File System (NFS) is a distributed file system protocol that allows a user on a client computer to access files over a network similar to how local storage is accessed. NFS enables file sharing between multiple systems in a network, allowing users to work with shared directories as if they are local.

The NFS server hosts the shared directory and manages access permissions, while client machines mount the shared directories over the network.

Requirements

- Ubuntu system for NFS server setup.
- Client systems running Ubuntu or Windows.
- Sudo (administrative) privileges.
- Network connectivity between server and clients.

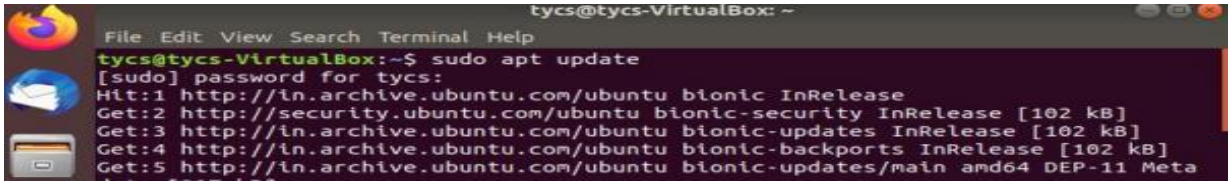
Step-by-Step Procedure

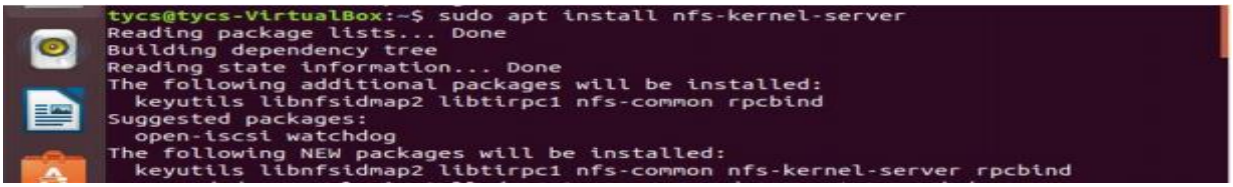
Step-1: Update package list and install NFS server

sudo apt update

sudo apt install nfs-kernel-server

ifconfig





- Check the network interface and IP address for client configuration.

Step-2: Create shared directory and add test file

pwd

mkdir NFSMain

cd NFSMain

nano Jayad

```
fycs@fycs-VirtualBox:~$ mkdir NFSMain
fycs@fycs-VirtualBox:~$ cd NFSMain
fycs@fycs-VirtualBox:~/NFSMain$ nano Jayad
```

- Inside nano, enter the message:

Hello Everyone!!

```
fycs@fycs-VirtualBox: ~/NFSMain
  GNU nano 2.5.3              File: Jayad                    Modified

hello everyone

                              [ New File ]
^G Get Help   ^O Write Out  ^W Where Is   ^K Cut Text   ^J Justify    ^C Cur Pos
^X Exit       ^R Read File  ^\ Replace    ^U Uncut Text^T To Spell   ^_ Go To Line
```

- Save and exit nano, then list files:

Ls

- Return to parent directory:

cd ..

```
fycs@fycs-VirtualBox:~/NFSMain$ ls
Jayad
fycs@fycs-VirtualBox:~/NFSMain$ cd..
cd..: command not found
fycs@fycs-VirtualBox:~/NFSMain$ cd
```

Step-3: Set directory permissions and configure exports

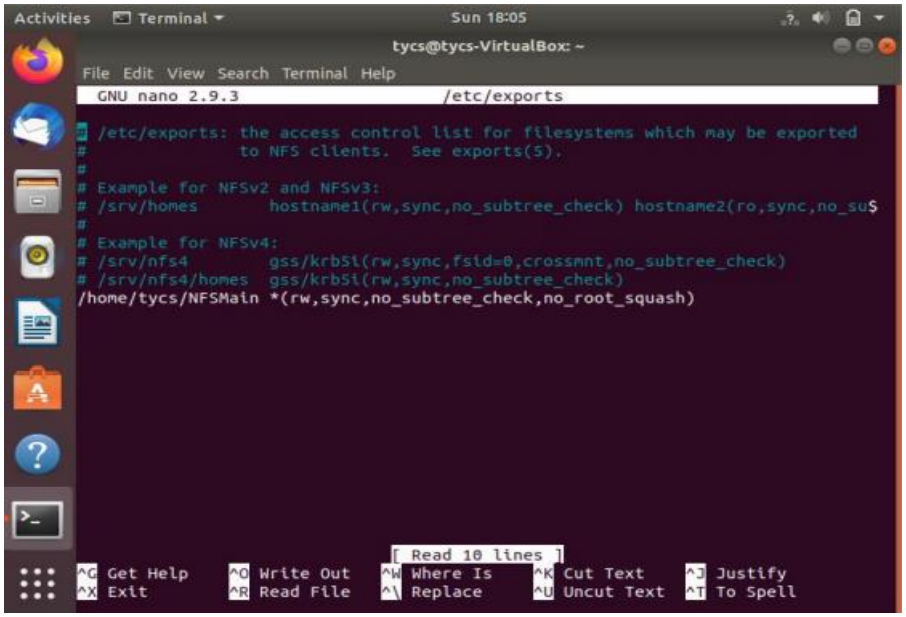
chmod 777 -R NFSMain

sudo nano /etc/exports

```
tycs@tycs-VirtualBox:~/NFSMain$ cd ..
tycs@tycs-VirtualBox:~$ chmod 777 -R NFSMain
tycs@tycs-VirtualBox:~$ sudo nano /etc/exports
```

- At the end of /etc/exports, add the line (adjust path and client network as needed):

/home/tycs/NFSMain *(rw,sync,no_subtree_check,no_root_squash)



Step-4: Export shared directories and manage NFS server service

sudo exportfs -a

showmount -e 10.0.2.15

sudo systemctl enable nfs-kernel-server

sudo systemctl start nfs-kernel-server

sudo systemctl status nfs-kernel-server

```
tycs@tycs-VirtualBox:~$ sudo exportfs -a
tycs@tycs-VirtualBox:~$ showmount -e 10.0.2.15
Export list for 10.0.2.15:
/home/mohsin/NFSMain *
/home/tycs/NFSMain   *
tycs@tycs-VirtualBox:~$ sudo systemctl enable nfs-kernel-server
Synchronizing state of nfs-kernel-server.service with SysV service script with
/lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable nfs-kernel-server
```

```
tycs@tycs-VirtualBox:~$ sudo systemctl start nfs-kernel-server
tycs@tycs-VirtualBox:~$ sudo systemctl status nfs-kernel-server
● nfs-server.service - NFS server and services
   Loaded: loaded (/lib/systemd/system/nfs-server.service; enabled; vendor pres
   Active: active (exited) since Sun 2025-08-03 17:52:00 IST; 6min ago
 Main PID: 4501 (code=exited, status=0/SUCCESS)
    Tasks: 0 (limit: 3558)
   CGroup: /system.slice/nfs-server.service

Aug 03 17:51:59 tycs-VirtualBox systemd[1]: Starting NFS server and services...
Aug 03 17:52:00 tycs-VirtualBox systemd[1]: Started NFS server and services.
tycs@tycs-VirtualBox:~$
```

- showmount -e <server_IP> verifies exported directories are shared to clients.

Observation / Output

- NFS server installed and running successfully.
- Directory /home/tycs/NFSMain is shared with full read-write permissions.
- Clients can view and access exported shares using the server IP.
- NFS service is enabled and active.

Conclusion

The NFS server was configured to share a directory on the Ubuntu system. The shared directory is accessible to NFS clients over the network, enabling centralized file sharing between Linux and Windows machines.

Troubleshooting / Notes

- Ensure the server IP (10.0.2.15) matches your actual server IP from ifconfig.
- For Windows clients, install and enable "Services for NFS" or use third-party NFS clients.
- Check firewall settings to allow NFS traffic (ports 2049 TCP/UDP).
- If clients face permission issues, verify chmod permissions and the no_root_squash option.

Practical No. 7 – Install MySQL & phpMyAdmin for Database Management

Aim

- To install and configure MySQL as a database server on Ubuntu.
- To install phpMyAdmin to manage MySQL databases via a web browser from client systems.

Theory

MySQL is an open-source relational database management system (RDBMS) used for data storage, retrieval, and management. phpMyAdmin is a web-based tool that provides a user-friendly graphical interface to administer MySQL databases, allowing tasks such as table creation, data insertion, and query execution from any web client.

Requirements

- Ubuntu system with internet.
- Sudo (admin) privileges.
- Web browser available on server or client systems.

Step-by-Step Procedure

Step-1: Install MySQL Server, Client, phpMyAdmin, and Web Components

sudo apt update

```
fycs@fycs-VirtualBox:~$ sudo apt update
[sudo] password for fycs:
Get:1 http://security.ubuntu.com/ubuntu xenial-security InRelease [106 kB]
Hit:2 http://in.archive.ubuntu.com/ubuntu xenial InRelease
Get:3 http://in.archive.ubuntu.com/ubuntu xenial-updates InRelease [106 kB]
Get:4 http://security.ubuntu.com/ubuntu xenial-security/main amd64 DEP-11 Metada
```

sudo apt install mysql-server

```
fycs@fycs-VirtualBox:~$ sudo apt install mysql-server
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  libaio1 libhtml-template-perl mysql-client-5.7 mysql-client-core-5.7
  mysql-common mysql-server-5.7 mysql-server-core-5.7
```

sudo apt install mysql-client

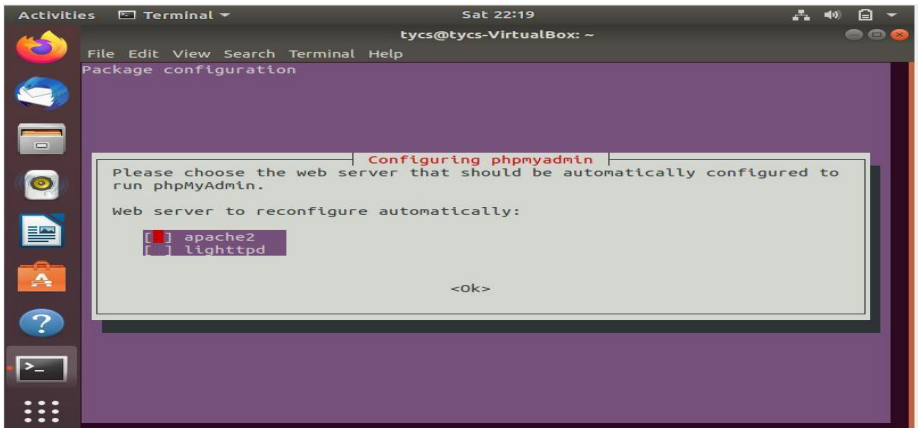
```
fycs@fycs-VirtualBox:~$ sudo apt install mysql-client
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
  mysql-client
0 upgraded, 1 newly installed, 0 to remove and 167 not upgraded.
```

sudo apt install phpMyAdmin

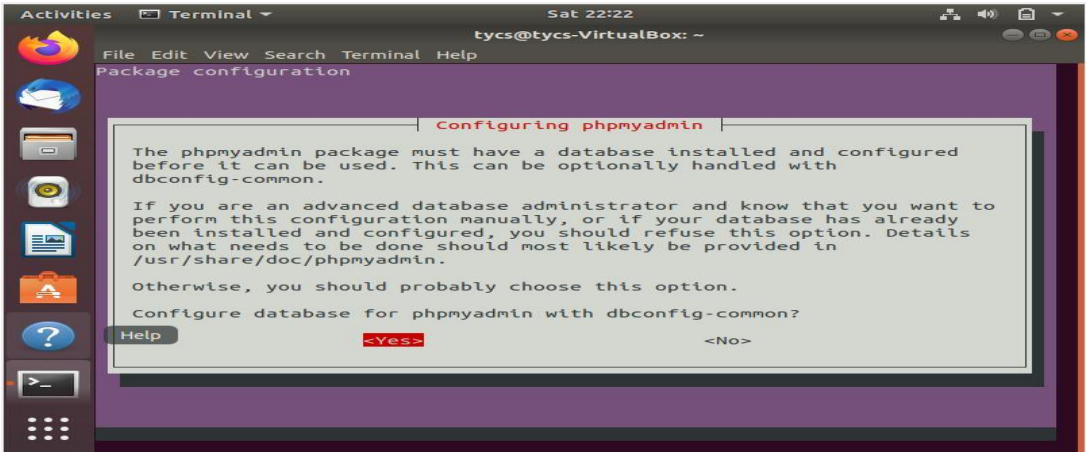
```
fycs@fycs-VirtualBox:~$ sudo apt install phpmyadmin
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils dbconfig-common
  dbconfig-mysql javascript-common libapache2-mod-php libapache2-mod-php7.0
  libapr1 libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap libjs-jquery
  libjs-sphinxdoc libjs-underscore liblua5.1-0 libmcrypt4 php-common php-gd
  php-gettext php-mbstring php-mcrypt php-mysql php-pear php-phpseclib
```

During installation:

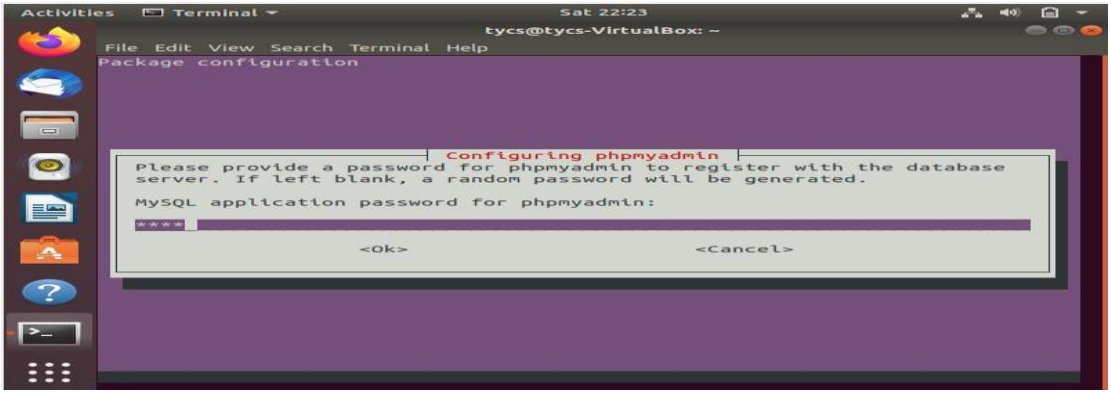
- Select "apache2" and press Enter



- Choose "Yes" to configure database for phpMyAdmin



- Set and confirm MySQL application password (example: 1234)



sudo apt install apache2

```
fycs@fycs-VirtualBox:~$ sudo apt install apache2
Reading package lists... Done
Building dependency tree
Reading state information... Done
apache2 is already the newest version (2.4.18-2ubuntu3.17).
apache2 set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 167 not upgraded.
```

sudo apt install php

```
fycs@fycs-VirtualBox:~$ sudo apt install php
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  php7.0
The following NEW packages will be installed:
```

Step-2: Enable and Start Services, Access MySQL

sudo systemctl enable mysql

sudo systemctl start mysql

sudo systemctl enable apache2

sudo systemctl start apache2

sudo mysql -u root -p

Enter root password (set during installation)

```
fycs@fycs-VirtualBox:~$ sudo systemctl enable mysql
Synchronizing state of mysql.service with SysV init with /lib/systemd/systemd-sy
sv-install...
Executing /lib/systemd/systemd-sysv-install enable mysql
fycs@fycs-VirtualBox:~$ sudo systemctl start mysql
fycs@fycs-VirtualBox:~$ sudo systemctl start apache2
fycs@fycs-VirtualBox:~$ sudo systemctl enable apache2
apache2.service is not a native service, redirecting to systemd-sysv-install
Executing /lib/systemd/systemd-sysv-install enable apache2
fycs@fycs-VirtualBox:~$ sudo systemctl start apache2
fycs@fycs-VirtualBox:~$ sudo mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 13
Server version: 5.7.33-0ubuntu0.16.04.1 (Ubuntu)
```

Step-3: Create Database, Table, and Insert Data in MySQL Shell

```
mysql> create database Jayad;
Query OK, 1 row affected (0.01 sec)

mysql> use Jayad
Database changed
mysql> create table tycs (roll_no int, name varchar(20));
Query OK, 0 rows affected (0.04 sec)

mysql> insert into tycs values(6091, 'Jayad');
Query OK, 1 row affected (0.03 sec)

mysql> select * from tycs
    -> ^C
mysql> select * from tycs;
+---------+-------+
| roll_no | name  |
+---------+-------+
|    6091 | Jayad |
+---------+-------+
1 row in set (0.00 sec)
```

Observation / Output

- MySQL server and client installed and running.
- phpMyAdmin installed and accessible via web browser.
- Successfully created a database, table, and inserted data.
- Verified data in the table using a SELECT statement.

Conclusion

The MySQL server is operational and accessible from the command line. phpMyAdmin allows browser-based database management, and the practical tasks (database/table creation, data insert, data view) were performed successfully.

Troubleshooting / Notes

- If phpMyAdmin is not found: Ensure Apache2 and PHP are properly installed, and phpMyAdmin is linked in /etc/apache2/conf-enabled/.

Practical No. 8 – Install and Configure Samba for File Sharing Between Linux and Windows

Aim

- To install and configure Samba on Ubuntu for sharing files/folders with Windows clients.
- To test the connection and access shared data from a Windows system.

Theory

Samba is an open-source software suite that allows file and print sharing between Unix/Linux and Windows systems using the SMB/CIFS networking protocol.

- It enables Linux systems to appear on a Windows network and share resources seamlessly.
- Windows clients can connect to Samba shares via the \\<IP> or network browser.
- Samba uses its own user authentication (smbpasswd) separate from Linux user accounts.

Requirements

- Ubuntu system with internet connection.
- Sudo (admin) privileges.
- A Windows client on the same network.
- Basic understanding of Linux commands.

Step-by-Step Procedure

Step 1: Update and Prepare Shared Folder

```
sudo apt update
mkdir tyshare
cd tyshare
sudo nano win.txt
cd ..
ls -l tyshare
sudo chmod -R 777 tyshare
ls -l tyshare
```

```
fycs@fycs-VirtualBox:~$ mkdir tyshare
fycs@fycs-VirtualBox:~$ cd tyshare
fycs@fycs-VirtualBox:~/tyshare$ sudo nano J1.txt
fycs@fycs-VirtualBox:~/tyshare$ cd ..
fycs@fycs-VirtualBox:~$ ls -l tyshare
total 4
-rw-r--r-- 1 root root 82 Aug 15 16:46 J1.txt
fycs@fycs-VirtualBox:~$ sudo chmod -R 777 tyshare
fycs@fycs-VirtualBox:~$ ls -l tyshare
total 4
-rwxrwxrwx 1 root root 82 Aug 15 16:46 J1.txt
```

- Creates the shared folder tyshare, adds a file win.txt, and sets open permissions.

Step 2: Set Ownership, Create Samba User, and Configure Share

sudo chown -R tycs:tycs tyshare

ls -l tyshare

uname

sudo useradd Abdullah

sudo smbpasswd -a Abdullah

sudo nano /etc/samba/smb.conf

```
fycs@fycs-VirtualBox:~$ sudo chown -R fycs:fycs tyshare
fycs@fycs-VirtualBox:~$ ls -l tyshare
total 4
-rwxrwxrwx 1 fycs fycs 82 Aug 15 16:46 J1.txt
fycs@fycs-VirtualBox:~$ uname
Linux
fycs@fycs-VirtualBox:~$ sudo useradd Jayad
fycs@fycs-VirtualBox:~$ sudo smbpasswd -a Jayad
New SMB password:
Retype new SMB password:
Added user Jayad.
fycs@fycs-VirtualBox:~$ sudo nano /etc/samba/smb.conf
```

```
fycs@fycs-VirtualBox: ~/tyshare
  GNU nano 2.5.3              File: J1.txt                    Modified

this file was created by the head of the table(Jayad) and is displayed in windo$

^G Get Help  ^O Write Out ^W Where Is  ^K Cut Text  ^J Justify   ^C Cur Pos
^X Exit      ^R Read File ^\ Replace   ^U Uncut Text^T To Spell  ^_ Go To Line
```

In the [smb.conf] file, add at the end:

```
  GNU nano 2.5.3            File: /etc/samba/smb.conf

    browseable = yes
    read only = yes
    guest ok = no
# Uncomment to allow remote administration of Windows print drivers.
# You may need to replace 'lpadmin' with the name of the group your
# admin users are members of.
# Please note that you also need to set appropriate Unix permissions
# to the drivers directory for these users to have write rights in it
;    write list = root, @lpadmin

[tyshare]
path =/home/FYCS/tyshare
browseable = yes
valid user = Jayad
write list = Jayad

^G Get Help  ^O Write Out ^W Where Is  ^K Cut Text  ^J Justify   ^C Cur Pos
^X Exit      ^R Read File ^\ Replace   ^U Uncut Text^T To Spell  ^_ Go To Line
```

- Restart and check Samba service:

sudo systemctl restart smbd

sudo systemctl status smbd

```
fycs@fycs-VirtualBox:~$ sudo systemctl restart smbd
fycs@fycs-VirtualBox:~$ sudo systemctl status smbd
● smbd.service - LSB: start Samba SMB/CIFS daemon (smbd)
   Loaded: loaded (/etc/init.d/smbd; bad; vendor preset: enabled)
   Active: active (running) since Fri 2025-08-15 16:53:06 IST; 13s ago
     Docs: man:systemd-sysv-generator(8)
  Process: 4388 ExecStop=/etc/init.d/smbd stop (code=exited, status=0/SUCCESS)
  Process: 4399 ExecStart=/etc/init.d/smbd start (code=exited, status=0/SUCCESS)
```

Step 3: Disable Firewall for Testing

sudo systemctl stop ufw

sudo systemctl status ufw

sudo testparm

```
fycs@fycs-VirtualBox:~$ sudo systemctl stop ufw
fycs@fycs-VirtualBox:~$ sudo systemctl status ufw
● ufw.service - Uncomplicated firewall
   Loaded: loaded (/lib/systemd/system/ufw.service; enabled; vendor preset: enab
   Active: inactive (dead) since Fri 2025-08-15 17:02:07 IST; 6s ago
  Process: 4518 ExecStop=/lib/ufw/ufw-init stop (code=exited, status=0/SUCCESS)
 Main PID: 213 (code=exited, status=0/SUCCESS)
```

```
fycs@fycs-VirtualBox:~$ sudo testparm
Load smb config files from /etc/samba/smb.conf
rlimit_max: increasing rlimit_max (1024) to minimum Windows limit (16384)
WARNING: The "syslog" option is deprecated
Processing section "[printers]"
Processing section "[print$]"
Processing section "[tyshare]"
Unknown parameter encountered: "valid user"
Ignoring unknown parameter "valid user"
Loaded services file OK.
Server role: ROLE_STANDALONE
```

- Ensures firewall does not block Samba during testing.

Step 4: Check Network IP

```
fycs@fycs-VirtualBox:~$ ifconfig
enp0s3    Link encap:Ethernet  HWaddr 08:00:27:86:20:b1
          inet addr:10.0.2.15  Bcast:10.0.2.255  Mask:255.255.255.0
          inet6 addr: fe80::28b:9420:82eb:aed2/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:7090 errors:0 dropped:0 overruns:0 frame:0
          TX packets:2179 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:9292958 (9.2 MB)  TX bytes:165419 (165.4 KB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:1428 errors:0 dropped:0 overruns:0 frame:0
          TX packets:1428 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:129892 (129.8 KB)  TX bytes:129892 (129.8 KB)
```

- Note the server's IP address.

Step 5: Access the Share from Windows

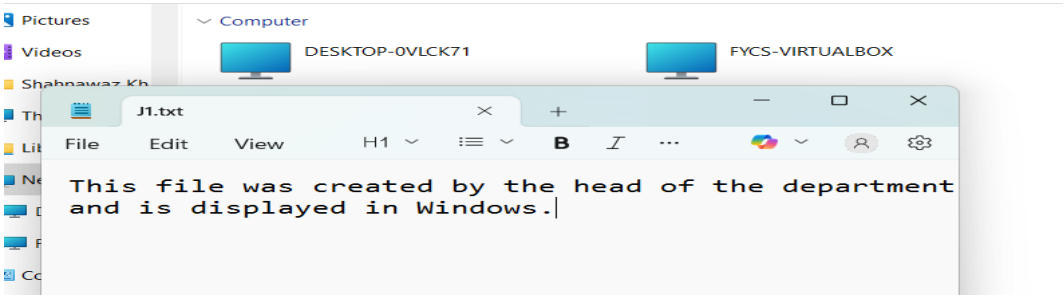
- On the Windows client:
    - Press Windows + R, type:

text

\\10.0.2.15

(replace with your Linux server's actual IP)

- Press Enter
- Open the folder tyshare
- Enter credentials:
  - Username: Jayad
  - Password: (set in Step 2)
- Access the shared Linux file: J1.txt



Observation

- Samba server installed and configured.
- Folder tyshare with file win.txt is accessible from Windows.
- User authentication successful for the Samba share.

Conclusion

Samba was successfully installed and setup on Ubuntu for cross-platform file sharing. The shared folder was accessible from a Windows client, allowing seamless document exchange and collaboration.

Troubleshooting / Notes

- If Windows client cannot connect, verify:
  - Server IP address and network connectivity.
  - User credentials match those created in Samba (smbpasswd).
  - Firewall is disabled or properly configured (sudo ufw allow samba).
- Use testparm to check Samba configuration syntax.

Practical No. 9 – LDAP Server Configuration

Part A: Configure LDAP Server and User Accounts
Part B: Install phpLDAPadmin Web Management Tool

Aim

- A: To install and configure LDAP server for centralized user account management on the local network, and add LDAP user accounts.
- B: To install and configure phpLDAPadmin for managing LDAP server through a web browser.

Theory

LDAP (Lightweight Directory Access Protocol) provides centralized authentication, storing and managing user accounts and directory information in a structured format.
OpenLDAP is the most widely used open-source LDAP implementation in Linux, mainly for enterprise identity management, network authentication, and user directory services.
phpLDAPadmin is a web-based tool to manage LDAP directories visually, making administrative tasks easier.

Requirements

- Ubuntu system with sudo privileges and internet access.
- Basic networking knowledge.
- Apache web server (for web interface).
- LDAP client utilities.

Part A – Configure LDAP Server, Add LDAP User Accounts

Step-by-Step Procedure

Step 1 – Update package list

sudo apt-get update

```
fycs@fycs-VirtualBox:~$ sudo apt-get update
[sudo] password for fycs:
Get:1 http://security.ubuntu.com/ubuntu xenial-security InRelease [106 kB]
Hit:2 http://in.archive.ubuntu.com/ubuntu xenial InRelease
Get:3 http://in.archive.ubuntu.com/ubuntu xenial-updates InRelease [106 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu xenial-backports InRelease [106 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu xenial-updates/main amd64 DEP-11 Metad
ata [314 kB]
Get:6 http://security.ubuntu.com/ubuntu xenial-security/main amd64 DEP-11 Metada
```

Step 2 – Install LDAP server and utilities

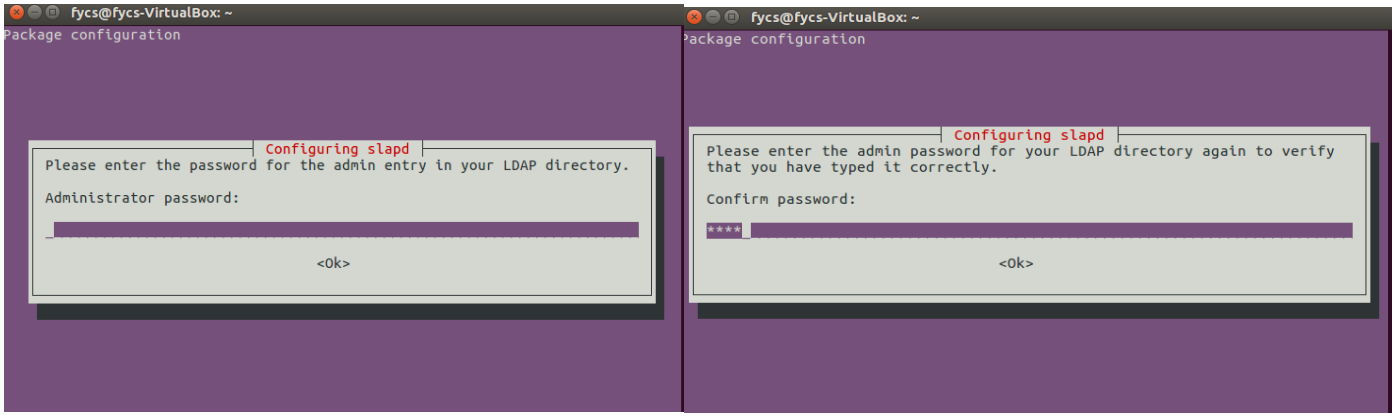
sudo apt-get install slapd ldap-utils

```
fycs@fycs-VirtualBox:~$ sudo apt-get install slapd ldap-utils
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  libodbc1 libslp1
```

Step 3 – Configure LDAP server settings

sudo dpkg-reconfigure slapd

```
fycs@fycs-VirtualBox:~$ sudo dpkg-reconfigure slapd
  Omitting slapd configuration as requested.
```



- Set your domain suffix (e.g., dc=example,dc=com), organization name, and admin password through interactive prompts.

Step 4 – Check slapd service status

```
fycs@fycs-VirtualBox:~$ sudo systemctl status slapd
● slapd.service - LSB: OpenLDAP standalone server (Lightweight Directory Access
   Loaded: loaded (/etc/init.d/slapd; bad; vendor preset: enabled)
  Drop-In: /lib/systemd/system/slapd.service.d
           └─slapd-remain-after-exit.conf
   Active: active (running) since Fri 2025-08-15 18:22:32 IST; 22s ago
     Docs: man:systemd-sysv-generator(8)
```

- Ensure the LDAP server is running.

Step 5 – Verify LDAP server listens on port 389

```
fycs@fycs-VirtualBox:~$ sudo lsof -i :389
COMMAND  PID     USER   FD   TYPE DEVICE SIZE/OFF NODE NAME
slapd   4187 openldap    9u  IPv4  34926      0t0  TCP *:ldap (LISTEN)
slapd   4187 openldap   10u  IPv6  34927      0t0  TCP *:ldap (LISTEN)
```

Step 6 – Identify current LDAP identity

ldapwhoami -x -H ldap://localhost:389

```
fycs@fycs-VirtualBox:~$ ldapwhoami -x -H ldap://localhost:389
anonymous
```

Step 7 – Query LDAP directory base

ldapsearch -x -H ldap://localhost:389 -b "dc=example,dc=com"

```
fycs@fycs-VirtualBox:~$ ldapsearch -x -H ldap://localhost:389 -b "dc=example,dc=
com"
# extended LDIF
#
# LDAPv3
# base <dc=example,dc=com> with scope subtree
```

Observation

- LDAP (OpenLDAP) service installed, started, and verified.
- Server responds to queries and is accessible over network port 389.
- Directory base is reachable.

Conclusion

OpenLDAP server was successfully installed, configured, and verified on the Ubuntu system. The service is ready to store and manage network user accounts.

Notes / Troubleshooting

- If slapd fails to start, review /var/log/syslog for errors.
- Use correct domain suffix during reconfiguration.

Part B – Install and Configure phpLDAPadmin Web Interface

Step-by-Step Procedure

Step 1 – Install phpLDAPadmin package

sudo apt-get install phpldapadmin

```
fycs@fycs-VirtualBox:~$ sudo apt-get install phpldapadmin
[sudo] password for fycs:
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  php-ldap php7.0-ldap
The following NEW packages will be installed:
  php-ldap php7.0-ldap phpldapadmin
0 upgraded, 3 newly installed, 0 to remove and 163 not upgraded.
Need to get 750 kB of archives.
```

Step 2 – Configure phpLDAPadmin

sudo nano /etc/phpldapadmin/config.php

```
fycs@fycs-VirtualBox:~$ sudo nano /etc/phpldapadmin/config.php
fycs@fycs-VirtualBox:~$
```

```
  GNU nano 2.9.3              /etc/phpldapadmin/config.php

    'ldap.example.com',
    'ldaps://ldap.example.com/',
    'ldapi://%2fusr%local%2fvar%2frun%2fldapi'
            (Unix socket at /usr/local/var/run/ldap) */
$servers->setValue('server','host','127.0.0.1');

/* The port your LDAP server listens on (no quotes). 389 is standard. */
// $servers->setValue('server','port',389);

/* Array of base DNs of your LDAP server. Leave this blank to have phpLDAPadmin
   auto-detect it for you. */
$servers->setValue('server','base',array('dc=example,dc=com'));

/* Five options for auth_type:
   1. 'cookie': you will login via a web form, and a client-side cookie will
      store your login dn and password.
   2. 'session': same as cookie but your login dn and password are stored on t$
      web server in a persistent session variable.
   3. 'http': same as session but your login dn and password are retrieved via
      HTTP authentication.
   4. 'config': specify your login dn and password here in this config file. No
      login will be required to use phpLDAPadmin for this server.
   5. 'sasl': login will be taken from the webserver's kerberos authentication.
```

- Adjust server connection settings if necessary ($servers->setValue('server','host','127.0.0.1'); etc.)

Step 3 – Edit Apache config for phpLDAPadmin

sudo nano /etc/apache2/conf-enabled/phpldapadmin.conf

```
fycs@fycs-VirtualBox:~$ sudo nano /etc/apache2/conf-enabled/phpldapadmin.conf
[sudo] password for fycs:
```

```
  GNU nano 2.9.3      /etc/apache2/conf-enabled/phpldapadmin.conf

# Define /phpldapadmin alias, this is the default
<IfModule mod_alias.c>
    Alias /phpldapadmin /usr/share/phpldapadmin/htdocs
</IfModule>

# You can also use phpLDAPadmin as a VirtualHost
# <VirtualHost *:*>
#     ServerName ldap.example.com
#     ServerAdmin root@example.com
#     DocumentRoot /usr/share/phpldapadmin
#     ErrorLog logs/ldap.example.com-error.log
#     CustomLog logs/ldap.example.com-access.log common
# </VirtualHost>

<Directory /usr/share/phpldapadmin/htdocs/>

    DirectoryIndex index.php
    Options +FollowSymLinks
    AllowOverride None

    Order allow,deny
    Allow from all

                                [ Read 51 lines ]
^G Get Help    ^O Write Out   ^W Where Is    ^K Cut Text    ^J Justify
^X Exit        ^R Read File   ^\ Replace     ^U Uncut Text  ^T To Spell
```

- Ensure proper Apache alias and access permissions.

Step 4 – Restart and check Apache web server

sudo systemctl restart apache2

sudo systemctl status apache2

```
fycs@fycs-VirtualBox:~$ sudo systemctl restart apache2
fycs@fycs-VirtualBox:~$ sudo systemctl status apache2
● apache2.service - LSB: Apache2 web server
   Loaded: loaded (/etc/init.d/apache2; bad; vendor preset: enabled)
  Drop-In: /lib/systemd/system/apache2.service.d
           └─apache2-systemd.conf
   Active: active (running) since Fri 2025-08-15 19:48:21 IST; 13s ago
     Docs: man:systemd-sysv-generator(8)
  Process: 5325 ExecStop=/etc/init.d/apache2 stop (code=exited, status=0/SUCCESS
  Process: 4836 ExecReload=/etc/init.d/apache2 reload (code=exited, status=0/SUC
  Process: 5347 ExecStart=/etc/init.d/apache2 start (code=exited, status=0/SUCCE
   CGroup: /system.slice/apache2.service
           ├─5365 /usr/sbin/apache2 -k start
           ├─5368 /usr/sbin/apache2 -k start
           ├─5369 /usr/sbin/apache2 -k start
           ├─5370 /usr/sbin/apache2 -k start
           ├─5371 /usr/sbin/apache2 -k start
           └─5372 /usr/sbin/apache2 -k start
```

Observation

- phpLDAPadmin installed and configured.
- Apache web service restarted; web interface should be available.

Conclusion

phpLDAPadmin was successfully installed, providing a web-based graphical management tool for LDAP server administration.

Notes / Troubleshooting

- Access phpLDAPadmin at:

  http://localhost/phpldapadmin

If unable to connect, make sure Apache is running and firewall allows HTTP (port 80):

sudo ufw allow 80

- Use LDAP admin credentials set during slapd configuration to log in.
- For external access, configure Apache to allow remote or set proper permissions in its config file.